



Synthesis and properties of polypropylene-graphene nanocomposite

Rishaban R., Balakrishnan G., Hema R. *

Department of Electronics and Communication Engineering, Bharath University, Affiliated to Bharath Institute of Higher Education and Research, Chennai, Tamil Nadu 600073, India

ARTICLE INFO

Article history:
Available online 1 March 2022

Keywords:
Graphene
Exfoliation
Ball milling
Polymer
Nanocomposite

ABSTRACT

Graphene being 2D crystalline material has attracted broad and deep interest all over the world due to its good physical, chemical, mechanical and electrical property with high shape asymmetry. Fabrication of high-quality graphene is becoming increasingly important to work on its various properties and application prospects. Recent progress on the hybrid material-based nanomaterial has greater prospects in many applications. In this investigation, commercial graphite powder is preferred as the precursor and two different solvents showing different exfoliation efficiencies, i.e., N-Methyl-2-pyrrolidone and 1-Methylnaphthalene are employed as exfoliation media. Ball milling technique, known to have successfully produced nanomaterials in large scale, is employed in this work to produce multi-layered graphene sheets. Multi-layer graphene's are made with high aspect ratio with modifications in process parameter. The exfoliated multi-layered graphene is incorporated as filler material with polypropylene matrix to develop polymer nanocomposites. The enhanced mechanical stability of the polypropylene/graphene nanocomposite and its respective morphology is studied in this work which has applications in photo-voltaics, electronic devices and facial reconstruction.

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1. Introduction

Graphene is a two-dimensional material with a flawless dispersion of carbon atoms in a hexagonal honeycomb shape, 0.1 nm between each atom, and a thickness of 0.3 nm [1]. Since the discovery of monolayer graphene in 2004, there has been a lot of progress, the material has been under intensive research owing to its fascinating combination of properties due to its high degree of anisotropy [2]. Though large-scale production is tedious, there are some new methods are adopted for monolayer graphene fabrication. Graphene tends to exhibit exciting mechanical properties due to its bonding nature and overlapping. The physical and mechanical property of graphene is good due to the C-C bonding in sp² bonds [3]. Geometry reconstruction of graphene sheets designed with folds also supports excellent mechanical performance such as enhanced tensile and shear strength and torsional rigidity. It is also considered to be one of the strongest materials

due to its strength almost 200 times greater than steel [4]. It also exhibits a high theoretical specific surface area (2630 m² g⁻¹), remarkable flexibility (Young's modulus ~ 1 TPa), high hardness (15–24 kg mm⁻²), and also high corrosion resistance. The electrical properties include a tenable bandgap, high mobility of charge carriers (10,000 cm² V⁻¹ s⁻¹) at room temperature, excellent thermal conductivity (5000 Wm⁻¹ K⁻¹ and 10 times higher than that of copper). Quantum Hall Effect is observed with the carriers at low temperature and high magnetic fields [5].

Obratsov et al. synthesized few-layer graphene sheets using the thermal CVD on a Ni substrate. Fabrication of the monolayer graphene on large scale is considered to be a challenge and many of the research groups are working on it [6]. Single layer to few layers was also grown on Ni substrate by some research groups. PECVD methods were also used for the single-layer and double-layer graphene fabrication. Some research groups have also reported the fabrication of high-quality graphene sheets using the thermal CVD process. Other methods for isolating single functionalized graphene sheets have also been reported, in which one of the most common and efficient methods is the thermal exfoliation of reduced graphene oxide (GO) which involved the oxidation and exfoliation of graphite [7]. Graphite is known to be a regularly

Abbreviations: CVD, Chemical Vapor Deposition; PECVD, Plasma enhanced Chemical Vapor Deposition.

* Corresponding author at: Bharath University, Affiliated to Bharath Institute of Higher Education and Research, Chennai, Tamil Nadu, India.

E-mail address: ingalgeri.bharathuniv@gmail.com (R. Hema).

<https://doi.org/10.1016/j.matpr.2022.02.206>

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stacked entity comprising graphene sheets held together by the van der Waals forces between the π - π stacked graphene layers, which have an interlayer spacing of 0.34 nm. Thus, the obtained GO sheet, decorated with immense oxygen-containing functional groups, can be good dispersant in aqueous environments are facilitative for further processing. However, this method too has its significant disadvantages, which includes the fact that the p-orbital structure is disrupted by oxidation, making GO a poor electrical conductor [8].

Alternative methodologies to overcome the above difficulty are reported, one of which is the production of graphene by directly exfoliating graphite by means of solvent assistance without oxidation. Stable suspension of graphene was achieved with solvent phase method [9]. There are two main approaches: sonication of graphite in solvents (e.g., natural graphite) and the ball milling technique (e.g., few- to multi-layered graphene), that employs exfoliation of graphene in solvent by zeroing the enormous van der Waals-like forces between graphite interlayer spacing which is greater than 5 Å [10]. Considering the factors such as energy consumption, time and the yield efficiency of graphene, the ball milling technique was comparatively much conservative than the sonication approach [11]. The sonication approach had a relatively low, typically $<0.01 \text{ mg mL}^{-1}$ yield efficiency and as to improve the graphene concentration, a long duration (4400 h) of sonication, involving longer power consumption, had to be employed. Application prospects of the single layer and multi layered graphene include electronic, mechanical, healthcare and many more. Recent progress on the hybrid material-based nanomaterials has greater prospects in applications such as photovoltaics, electronic devices and facial reconstruction.

2. Materials and methods

Synthetic graphite with a mean particle size of 4 μm and a density of about 2.3 g/cm^3 was used in the grinding experiments. The organic solvents are N-Methyl-2-pyrrolidone (NMP) and 1-Methylnaphthalene (Purity, 97.0%). Polypropylene matrix (PP1110 MG grade) was considered the polymer source. N-Methyl-2-pyrrolidone (NMP) and 1-Methylnaphthalene (NMP) are excellent solvents that are frequently utilised in industry and in the processing of nanomaterials. Despite its widespread use, it is the liquid structure of the substance that determines its solvation qualities. The qualities of these solvents, such as degree of order and interaction balance, justify NMP's high boiling point and varied ability to solvate both charged and uncharged species [12]. Additives (or Antioxidants) such as AO-168 [Tris (2,4-di-*tert*-butylphenyl) phosphite], AO-1010 [Tetrakis [methylene-3(3', 5'-di-*tert*-butyl-4'-hydroxyphenyl) propionate] methane] and calcium acetate (cast) were preferred in order to improve the processability of the composite. Finally, paraffin oil was obtained for better polymer-filler dispersion.

2.1. Exfoliation of multi-layered graphene

The milling experiments were carried out in the planetary ball mill unit (Retsch, PM 400, Latvia). Typically, the graphite materials with weight percentage of 3% and 5% were dispersed in the two different organic solvents, Methylnaphthalene and NMP with respect to a total volume (considered as 200 mL). The dispersed batch were added with wear resistant commercially available zirconium oxide milling beads with diameters 1 mm, 500 μm and 300 μm , transferred in the grinding jar (250 mL volume) and it was subjected to sequential grinding technique via ball milling. This technique provides a batch mode comminution of the product for the duration of 11 h (i.e., 3 h of milling with 1 mm, 4 h added

with 0.5 mm and 0.3 mm beads individually) with the rotational speed of the stirrer set to 275 rpm. As the approach involves the assistance of organic solvents, washing the samples after the milling process, was considered an important step to eliminate the organic compounds. Therefore, the obtained precipitates were subjected to centrifugation (Rotation: 6500 rpm; Time: 15 mins) and ultra-sonication (Amplitude: 40%; Time: 10 mins) dispersed in hexane, distilled water and isopropyl alcohol. The process was repeated 6 times and the final precipitates were dried at vacuum oven at 100 °C overnight.

2.2. Preparation of Pp/MLG nanocomposites

Graphene and graphite PP Nanocomposites (1% and 3% w/w) were prepared via 2 methodologies: Micro compounding and Injection moulding. The Masterbatch of the polymer nanocomposites were prepared as follows: As-synthesized MLG and pristine graphite (Source: Timrex cybrid), were added to the PP matrix. Detailed concentration of the masterbatch, in terms of wt%, and its corresponding sample identification is illustrated in Fig. 1.). In order to improve the processability and reduce the polymer degradation, additives such as AO-1010, AO-168 and calcium acetate at concentration of 300 ppm, 700 ppm and 70 ppm were added along with few drops of paraffin oil and was stirred well until a homogeneous dispersion was obtained. Then the Masterbatch with graphite and MLG as fillers were subjected to Micro compounding handled by Micro- compounder (HAAKETM MiniLab 2, Thermo scientific, USA), melted at set temperature (TM) of 260°C; and rotation speed of 151 min^{-1} . The obtained melt was then subjected to a very high pressure (10 MPa), temperature (260 °C) and molded via Injection molding (HAAKETM MiniJet 2, Thermo scientific, USA). [Supplementary Fig. S1](#) shows a typical Micro Compounding (A), Injection moulding (B) and ball mill unit (C).

3. Results and discussion

3.1. Raman spectroscopy

Raman spectroscopy is a powerful tool to determine the exact number of graphene layers. The D-peak ($\sim 1350 \text{ cm}^{-1}$) is detected for all the samples synthesized, owing to the breathing modes of sp^2 -bonded atoms, and reflects the disorder of the graphene layer. Raman spectra of delaminated sheets of different as-synthesized samples were measured and a considerable blue shift in the 2D band was observed, as shown in Fig. 2. This shift from 2679 cm^{-1} to 2674 cm^{-1} in the sample MLG-NMC from source graphite indicates the exfoliation of the graphite material. The same is observed in the sample MLG-NPK, i.e., from 2694 cm^{-1} to 2685 cm^{-1} .

The ratio of the D-peak intensity to the G-peak intensity ($I_D = I_G$) is often used as the measurement of the defect density of the graphene. As shown in Fig. 2, $I_D: I_G$ is ~ 0.4 , for all the exfoliated samples except for 0.85 for the MLG-NMC sample confirming that there are considerable defects in graphene. Also, $I_{2D}: I_G$ calculated to estimate the number of graphene layers is found to be ~ 0.5 for all the exfoliated samples, except for MLG-NMC sample valued, 0.9. These values indicate that the exfoliated samples are multi-layered. Thus, Raman spectra results the fact that the samples are well exfoliated. The band intensity ratios of the as-synthesized samples and its number of layers are calculated and is illustrated in Table 1.

3.2. FE-SEM studies

Scanning Electron Microscopy (SEM) for the MLG samples were performed and the morphology was studied, shown in Fig. 3. Fig-

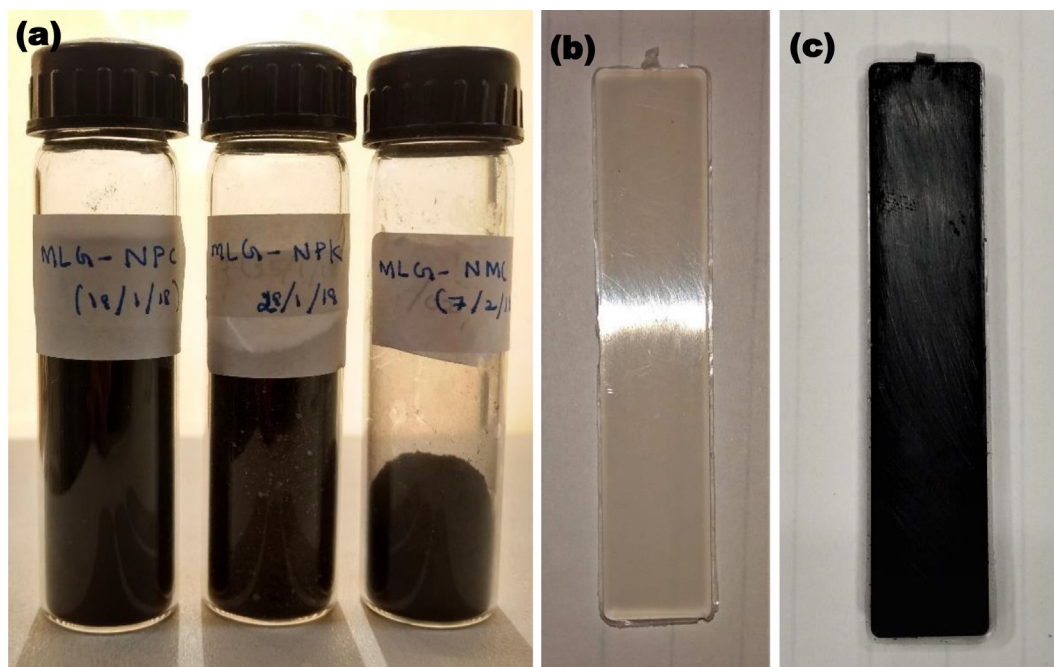


Fig. 1. (a) As-synthesized MLG and PP/MLG samples; (b) PP composite; (c) PP/MLG nanocomposite.

ure (A) and (F) shows SEM images of the graphite sources (CyPbrid and KS4) with the flake-like structure (and platelet-like structure under high magnification) having a small diameter of $\sim 1 \mu\text{m}$ and also some large flakes of about $\sim 5 \mu\text{m}$ in diameter. In contrast to the samples MLG-NPC, MLG-NMC and MLG-NPK, the small flakes are no more visible and the formation of few- to multi-layered graphene is observed. Figure (C), (E), (G) shows images of sheets of graphene being folded, which clearly indicates that the assistance of solvent has intercalated between the graphitic layers very well leading to formation of multi-layered graphene.

Also, the SEM images of the samples MLG-NPC and MLG-NPK shows the presence of partially transparent sheets of graphene suggesting very few layers of graphene which is oriented or pointed in the upward direction, which sums up to the evidence in the formation of graphene, as in Figure (B) and (H). In addition to this, sample MLG-NMC milled with assistance of solvent NMP shows excellent homogeneity of individual graphene layers lying flat on top of each other, shown in Figure (D), suggesting the possibility of good electrical contact between the sheets. The sample MLG-NMC having high surface area and excellent morphology was incorporated as nano-filler in polypropylene matrix and PP/MLG nanocomposite was fabricated. Figure (I), (J) and (K) shows the SEM images of PP neat, PP/CYPBRID: 1 and PP/MLG-NMC: 1 sample. From the above images, the composite prepared with MLG filler (K) is observed to have higher roughness in morphology as compared to that of the other two samples. From this variation in roughness, it might be evident that PP/MLG sample shows excellent properties such as enhanced mechanical and thermal stability.

3.3. DLS AND BET studies

The exfoliated samples were investigated with dynamic light scattering (DLS) to obtain an approximate measure of the lateral size distributions. The samples were dispersed in ethanol via ultra-sonication, transferred into cuvette, subjected to 2 measurements (100 runs per measurement) where the sedimentation of the particle was also studied by the reduction and broadening in peak intensity. Particle size of $500(\pm 50)$ nm for the samples MLG-NPC and MLG-NPK was observed, as shown in Fig. 4(A) and (C). This indicates that 1-Methylnaphthalene used as the exfoliation media has employed particle size reduction of the respective samples. Whereas the particle size of sample MLG-NMC though detected as $700(\pm 50)$ nm, showed excellent stability (Fig. 4B) compared to other samples. The corresponding sample identification and its respective lateral dimensions measured by DLS is illustrated in Table 2. Specific surface area was determined from the results of N_2 physisorption at 77 K with surface area analyzer (AutoSorb, Quantachrome instruments) using Brunauer–Emmett–Teller (BET) multipoint methods. Samples were subjected to degassing under N_2 flow for 3 h at 180°C . BET surface areas measured for MLG-NMC exfoliated with NMP solvent sample was observed as $773.8 (\pm 0.2) \text{ m}^2 \text{ g}^{-1}$ which was comparatively larger than the samples MLG-NPC and MLG-NPK exfoliated with 1-Methylnaphthalene solvent, i.e., $399.12 \text{ m}^2 \text{ g}^{-1}$ and $133.73 \text{ m}^2 \text{ g}^{-1}$, respectively. This drastic increase in the specific surface area of MLG samples indicates the change of graphitic phase to graphene layers. Thus, the MLG-NMC sample with improved surface area is considered

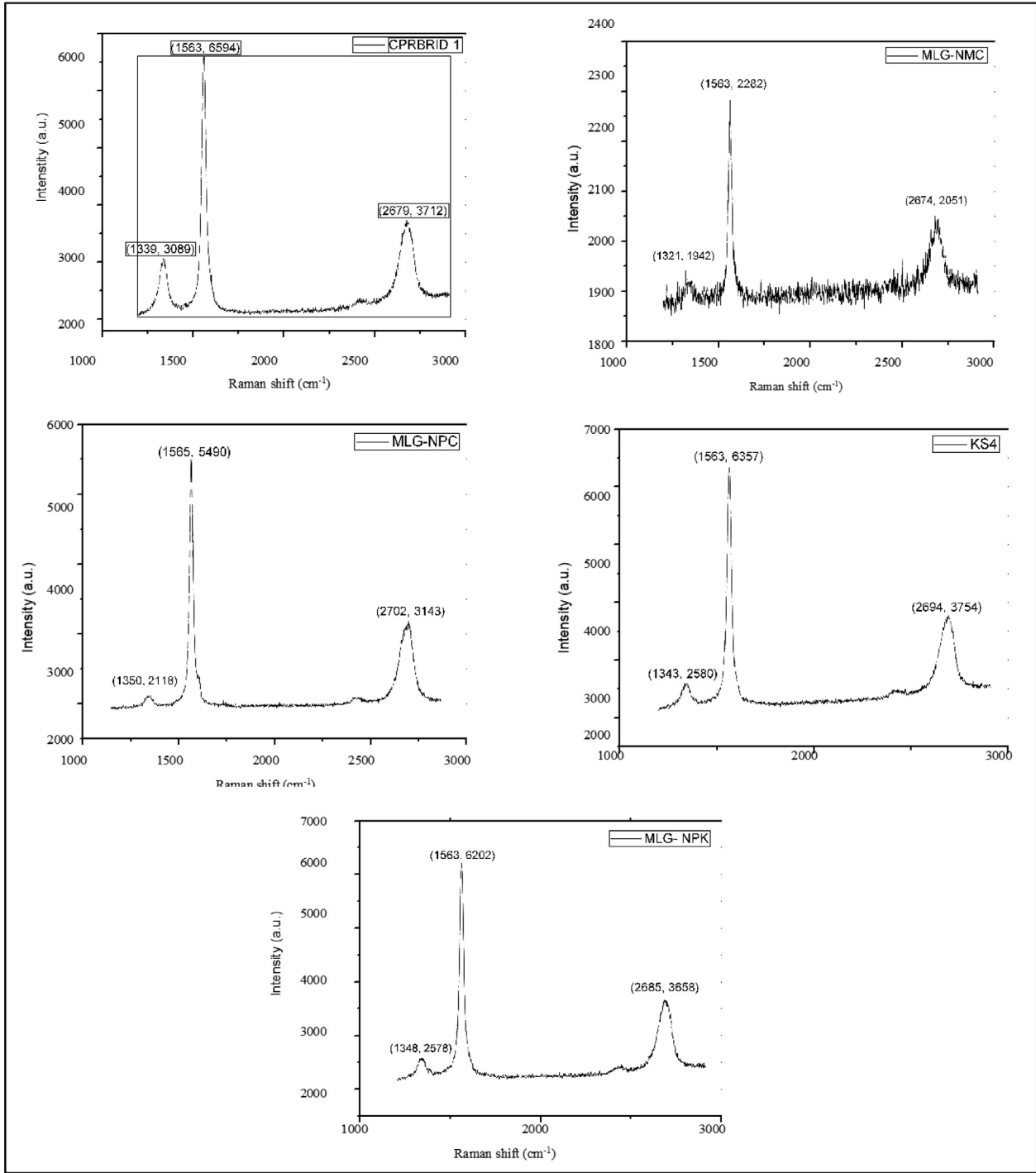


Fig. 2. Raman spectra for the corresponding as-synthesized samples.

Table 1
Intensity Ratio of 2D: G Peak and D: G Peak in Raman Spectra for the corresponding as-synthesized samples.

Sample Id	I _d /I _g ratio	I _{2d} /I _g ratio	No. of layers
CYPBRID	0.47	0.56	Multi-layered
KS4	0.41	0.58	Multi-layered
MLG- NPC	0.39	0.57	Multi-layered
MLG-NMC	0.85	0.90	Multi-layered
MLG-NPK	0.42	0.59	Multi-layered

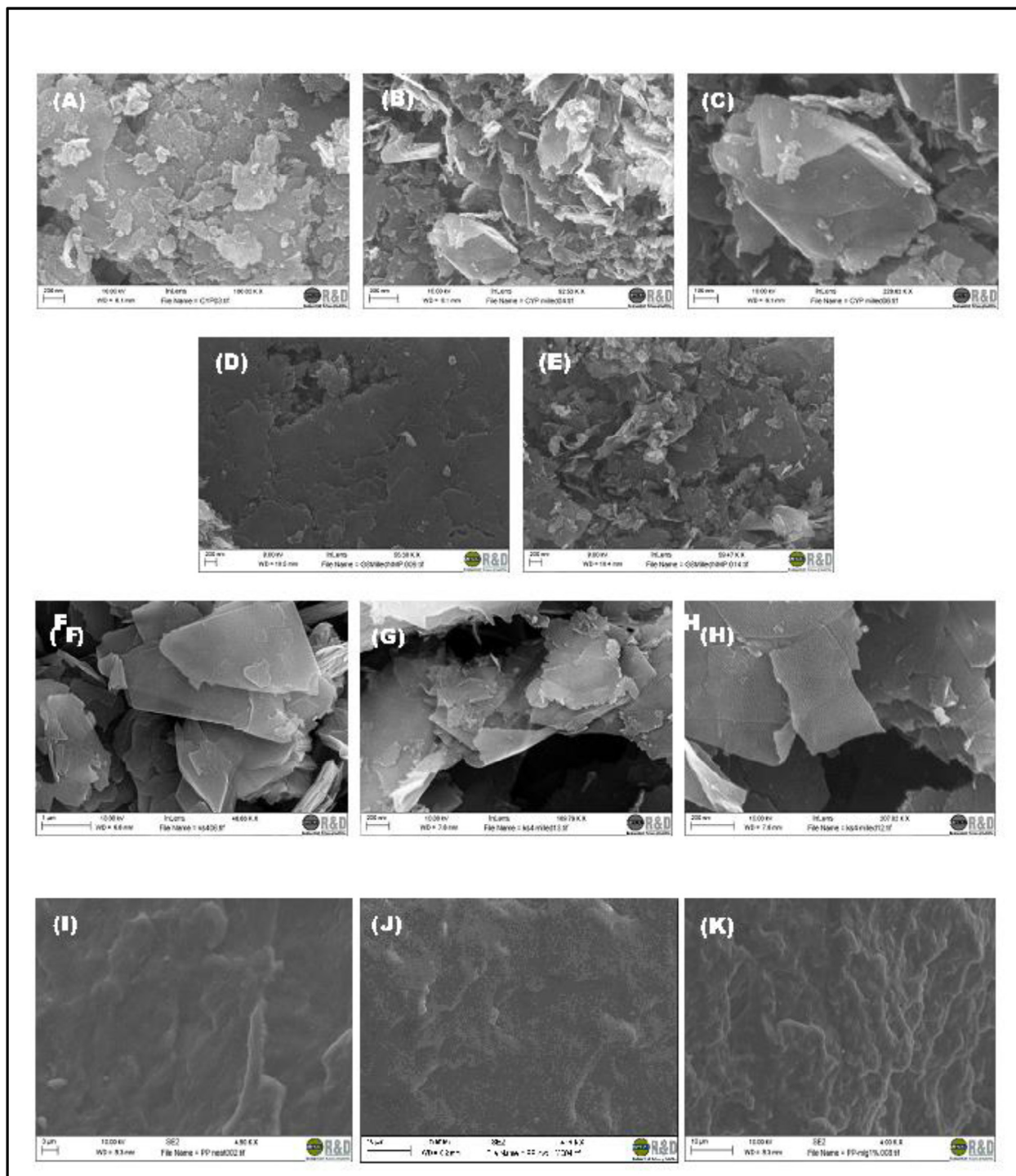


Fig. 3. SEM images of samples (A) cybrid graphite; (B) and (C) MLG-NPC; (D) and (E) MLG-NMC; (F) KS4 graphite; (G) and (H) MLG-NPK; (I) PP neat; (J) PP/CYPBRID:1; and (K) PP/MLG-NMC:1.

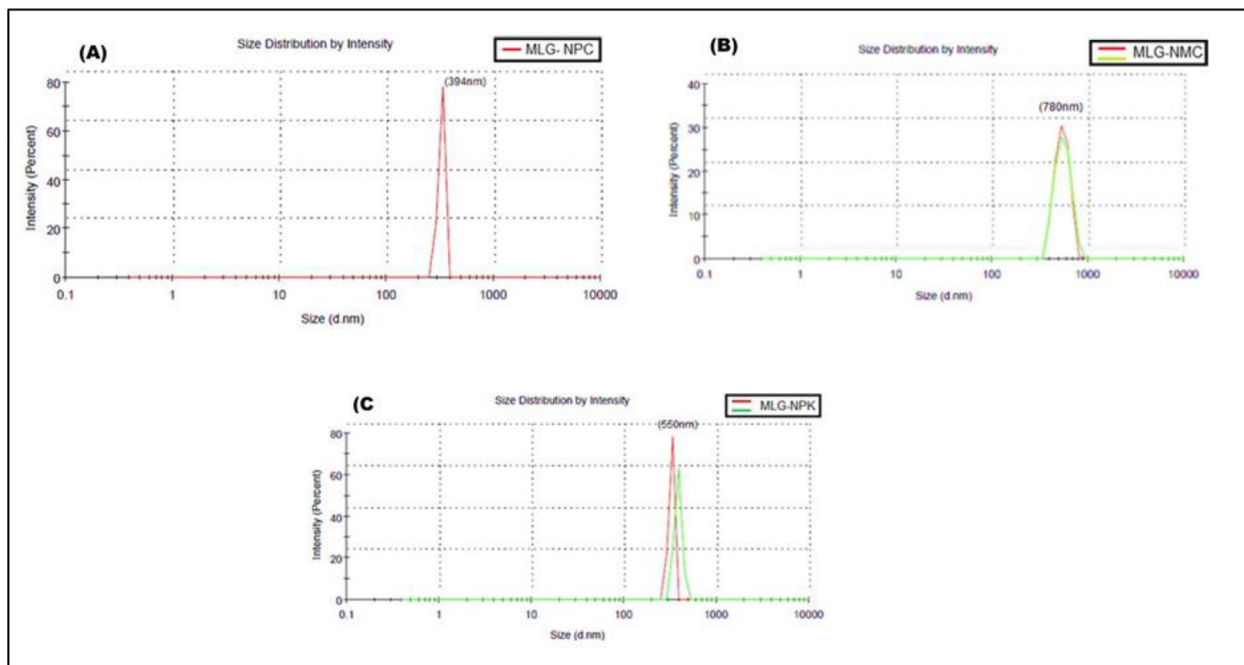


Fig. 4. Particle size distribution of the MLG samples synthesized. (A) MLG-NPC; (B) MLG-NMC; and (C) MLG-NPK.

Table 2

The corresponding sample identification and its respective specific surface areas and lateral dimensions.

Sample Id	Source graphite	Organic solvent used	Graphite(Wt%)	Surface area (M^2/G)	Lateral dimension
CYP.G	–	–	–	290.90	10 μm
KS4.G	–	–	–	90.14	5 μm
MLG-NPC	CyPbrid	1-Methylnaphthalene	3	399.12	400 nm
MLG-NMC	CyPbrid	N-Methyl-2-pyrrolidone	5	773.80	800 nm
MLG = NPK	KS4	1-Methylnaphthalene	5	133.73	600 nm

the perfect candidate as a filler material in the preparation of polymer nano-composites.

4. Conclusion

In the present work, multi-layered graphene material is successfully synthesized with an average lateral dimension of 500 nm via solvent-phase exfoliation of synthetic pristine graphite. The effect of addition of organic solvents N-Methyl-2-pyrrolidone and 1-Methylnaphthalene in the ball milling approach has been studied. The absence of harsh chemicals like strong acids, reducing agents, etc. makes this method much simpler and efficient. The morphology and thermal stability of the synthesized graphene nanosheets were investigated using Raman spectrometry, FE-SEM, DLS and surface area analyzer. The FE-SEM results indicates the formation of graphene layers. A lower 2D band shift in Raman spectra of the MLG samples confirms that the synthesized multi-layered graphene samples along with good thermal stability was preferred as the efficient nano-filler to fabricate the polymer/graphene nanocomposite. Thus, the polymer/multi-layered graphene nano-composites prepared has shown a remarkable enhancement in mechanical and thermal stability.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors wish to thank Bharath University, Chennai for their support and laboratory facilities to carry out the research work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matpr.2022.02.206>.

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